

LIBRARY SERVICES THROUGH MOBILE TECHNOLOGY: AN OVERVIEW

Issac Arputharaj J.^{1,a)} Sakthivel R.^{2,b)} Sivakumar, C.^{3,c)}

¹Assistant Professor, PG & Research Department of Library and Information Science, Bishop Heber College, Tiruchirappalli.²II M.Lib.I.Sc, PG & Research Department of Library and Information Science, Bishop Heber College, Tiruchirappalli.³II M.Lib.I.Sc, PG & Research Department of Library and Information Science, Bishop Heber College, Tiruchirappalli

^{a)}issac.li@bhc.edu.in

^{b)}ls211212004@bhc.edu.in

^{c)}sivakarthicksk22@qmac

om

ABSTRACT

Mobile phones and other handheld devices, such as I-pads, smart phones, e-readers, tablets, and so on, are examples of such devices that are primarily used for communication and multipurpose tasks such as voice recognition. Many people are unaware that these devices can be used for more than just communication, sending SMS messages, taking photos/videos, storing phone numbers, playing music, voice recording, and so on. The public library has undergone significant transformations, shifting from paper to electronic media. The proliferation of information in electronic media has had a significant impact on libraries. One aspect of electronic media is mobile technology. The primary goal of this paper is to examine how mobile technology is used in libraries, as well as the services provided by technology. In this quest, an attempt has been made to bring out the meaning of mobile libraries, types of library services provided through mobile technology, and a glimpse of trends, developments, and various projects undertaken by various leading libraries around the world, particularly in developed countries, as well as the role of vendors in this regard. The position of India in this scenario is also visible. This paper also attempts to understand how mobile technology can be used to improve and provide services.

INTRODUCTION

The foundation of the mobile library concept is the Internet, wireless networks, and multimedia technology. It offers access without limitations on time, place, or area through the use of numerous mobile devices. It is difficult to give users the relevant information at the appropriate time in the era of information explosion. Mobile technologies have enabled communication and information access for users from their own homes and offices, as well as from wherever they are on the go with their cellular phones or PDAs, very convenient and timely (personal digital assistants). With the advent of mobile technologies, the "Libraries in Hand" trend has emerged. Within a very brief amount of time, mobile devices can access information from a remote source.

These days, a lot of college and university libraries offer digital collections that may be viewed online. Mobile devices, on the other hand, offer a viable alternative for gaining access to digital libraries. On users' mobile phones, digital collections can be made accessible through a digital library. Mobile gadgets are truly personalized things. It is possible to provide more accurate, tailored information and services by using search histories and actual locations. The use of mobile phones in library services and the resulting difficulties are the main topics of this article.

The buildings where librarians work have been associated with them for generations. This concept is evolving quickly in the modern world as technology develops. Digital data is all around us. People can utilize it outside of libraries if they like. Information is accessible to them wherever they are, including at home, work, restaurants, and other public places. Libraries are able to offer their customers effective and quick library services with no barriers thanks to ICT.

In the past, the library communicated with its users through announcements and circulars, and they had to visit the library in person to acquire updates on its activities. However, in today's ICT environment, customers

can receive a single SMS on their mobile phones informing them of the library's activities. Libraries were forced to implement these adjustments in order to keep up with the changing times and to address the Customers want literature in a suitable format, which fits them the best.

LITERATURE REVIEW

Abdulqawiyyu Mustapha (et.al) (2021) highlighted Current trends in libraries especially in the academic library require remote and boundary less access to library services and services. Mobile Technology has been recognized as the simple, single most embraced technology in the world. Applying this technological innovation will bring a lot of relief to users in Bayero University, Kano library and the University community in general. The study investigates the preparedness of Bayero University, Kano library to apply mobile technology for information services delivery. The population consisted of 11 staff from the library's automation unit, with only two of them being sampled. The research instrument used was an interview. The study shows that the staff agree with the impact and relevancy of the application of the service in the library. The study also revealed challenge with the application of the service such as Lack of committed staff, lack of technical knowhow, poor of internet connection, lack of training. The study also recommends that the staff and Student at Bayero University, Kano Library should be educated about the benefits of using mobile-based library services for relevant information, Continuous training should be given to library staff in the application of mobile based- library services and Management should also develop strategy for the marketing of the mobile-based library services.

Hansaben Shashikantbhai Bhoj(2020) emphasized the application of mobile phone technology in all aspects of our everyday life cannot be ignored in institutions and establishments including libraries. Therefore libraries in developed economies have embraced the mobile phone revolution and are utilizing them for effective and efficient services. In contrast, however, mobile - based services are not pervasive among academic libraries in India. They are yet to be embraced by academic and research libraries. This paper reports findings of a survey on mobile - based library services in various libraries of India. The aim of the study was to seek the views of users regarding the use of mobile phones in library services. It sought to find whether they would be interested in using such service.

Satish Kumar (2015), Mobile phones and other handheld devices such as i-pads, smart phones, e-readers, tablets etc., are among such devices that are basically used for communication and performing multipurpose tasks such as voice communication, sending SMS'es, taking snapshots/videos, storing phone numbers, playing music, voice recording etc., many are not even aware that these devices can also be used for seeking information from libraries. Public library has seen tremendous changes and evaluated from paper to electronic media. Proliferation of information in electronic media found great impact on libraries. Mobile Technology is one of the aspects of Electronic Media. The main objective of this paper is to focus on how mobile technology is being used in libraries as well as services provided through technology. In this quest, an attempt has been made here to bring out the meaning of mobile libraries, types of library services provided through mobile technology and glimpse of trends, developments & various projects undertaken by different leading libraries across the globe, especially in the developed countries and also the role of vendors in this regard. India's position in this scenario is also seen. Also this paper is an attempt to understand how mobile technology is useful in improving and providing better library services.

Sagar Kumbhar and Rohan Pawar (2014) found that the development of Information and Communication Technology (ICT) have shifted the applications in library from the traditional to hybrid library, then automated library, digital archives stages, library 2.0 and mobile phone services. Nowadays Mobile phones have become the essential part of human life for communication and it also helps to student in e-learning. In this paper we have discussed about mobile technology and focused on its application and challenges in library services. With help of mobile devices, libraries can produce new services and provide faster access to its collection. The implication of mobile library services has created opportunities as well as challenges for academic libraries

WHAT ARE M-LIBRARIES?

"When people talk about mobile libraries, they generally mean a machine or truck outfitted as a ranging branch library." M- Libraries are shortened for " Mobile Libraries." This term is analogous to that of "Libraries on Wheels" services. However, it is a type of library service that, by leveraging the power of technology, delivers

Information and learning materials onto users' mobile or any hand held devices such as laptops, PDAs, palm top computers, smart phones, notebooks, and so on, which are almost ubiquitously available, anytime, anywhere by anyone and which expands the reach of the library.

HOW IT IS OPERATED?

The creation of a device support, device independent, common platform, in order to work with a wide range of devices, is the first step toward providing such services, which is done at the server end. It is critical to note that mobile library services will be supported by all browsers. The display should be able to fit into the small screen of mobile devices, allowing for easy communication with all mobile/handheld devices. WURFL (Wireless Universal Resource File) is an example of such software. It is a free and open source XML.

MOBILE ACTIVITIES

- E-mail
- Transfer file from one place to another via portable USB device
- Send and receive messages
- Download and read e-books

MOBILE TECHNOLOGY IN LIBRARIES

Libraries should investigate mobile devices as a means of connecting with users. Developing a mobile library website that allows users to access library hours, It is simpler than most people believe to search databases, etc.

- Mobile library content (books and articles) (E-books, e-articles, digital materials)
- Library services from mobile devices include inquiry support, resource discovery, circulation, statistics collection, and study carrel booking (enquiries, statistics, college management, and circulation).
- Mobile-specific contexts/services (location services, QR codes, augmented reality)

LIBRARY SERVICES WITH THE HELP OF MOBILE TECHNOLOGY

Many academic and special/research libraries that implement these changing technologies provide a variety of services to their patrons, increasing the potential of such ICT devices. The provision of various types of services via this device is dependent on the target, i.e., the specific users', but the most common services provided in general are

a. Mobile Library Website: This service allows users to access library services via the Internet. Accessible materials include MP3, video clips, e-books, journal articles, and general information.

b. Text Messaging Services: A growing number of libraries are utilizing text messaging services such as SMSes for renewals, reservations, recall notices, and information about the arrival of new books, among other things.

c. Mobile Catalogue Search: Some universities in Western countries offer OPAC services via mobile, allowing students to check the availability of documents from the library.

d. Reference Services: A traditional service provided by a reference librarian. Users can now be referred using mobile technology.

e. Mobile Catalogue (M-OPAC): A mobile online public access catalogue that allows users to access information about library holdings from any location and at any time.

f. Social Networking: Library services and updates via Facebook, Twitter, RSS feeds, blogging, and other platforms.

g. E-mail: Current Awareness Service (CAS) can be more effectively provided.

h. Mobile-friendly websites or mobile applications: Researchers want to learn about important websites in their field of study. Using mobile technology, a list of websites can be easily sent.

i. Mobile e-Books: Users can access e-books via mobile devices such as the iPhone, iPad, Smartphone and Amazon offers the Kindle e-book reader.

j. SMS Service: This service is especially beneficial to the circulation department.

Libraries are using SMS for

- If the book requested is available (collect messages).
- If a book is due, set a reminder.
- Requesting a loan list via SMS.
- Using SMS to renew books.
- Requesting a summary of outstanding fines via SMS.
- Using SMS to check the availability of books.
- Requesting the library's opening hours via SMS.
- Service "Ask a Librarian."
- Allow users to search the catalogue.
- SMS alert
- Database accesses
- Recommend e-Readers – e- Books audio books.

IMPACT ON LIBRARIES

People prefer receiving information via SMS, and it is more popular among library users. Many users prefer smartphones to browse the internet, watch videos, and view photos.

- Because information can be delivered quickly, users save time.
- It attracts people to the library.
- It benefits from teaching and learning activities, which improves cooperation between library staff and users.
- Messages should be delivered more actively to people wherever they are at the right time.

Benefits of using a mobile phone Services

- Interactive Features:
 - Users of the mobile Web can access information fast and communicate with library staff to get particular references when away from the library.
- Information Accessibility:
 - Users who are unable to physically visit a library would benefit greatly from the ability to access information through mobile web, which also acts as a constant link to the resources they need.
- Time Savings:
 - Users using mobile devices can browse and search library materials. Users save time since they don't have to wait in line at the library's checkout desk to renew or reserve books.
- Customized Services:
 - A customised and specialised service, mobile library service enables users to connect with librarians and find relevant references.
 - User Involvement: By enabling users to include user-generated information like notes or photographs provided by users, libraries may improve OPAC.
- Unfettered Access:
 - All online resources that they can access on their PC are also available on their mobile devices.

CHALLENGES IN USING MOBILE TECHNOLOGY

- It is costly and requires a lot of resources.

- It necessitates skilled and knowledgeable library personnel. Screen Quality - Because mobile phones have small screens and displays, it is difficult to read.
- Internet connectivity and mobile signals are significant impediments.
- It necessitates adequate infrastructure and upkeep.

MOBILE LIBRARY DEVELOPMENTS AROUND THE WORLD

Open University, UK: Keren Mills of the Open University in the United Kingdom worked on a project called "Information Use on the Move." Cambridge University has provided funding to identify the patrons seeking information are increasingly using mobile phones.

Cambridge University: Cambridge University is one of the institutions that has created a mobile-friendly website for their digital library. This university's mobile site focuses on allowing users to search the catalogue using thumbnail images of book covers and access their account to view profiles.

Amsterdam University Library: The interim mobile library website developed by Amsterdam University Library offers a variety of services such as access to the OPAC, a simple search interface, a link to the library's social networking page on Twitter, information on library opening hours, location, contact information, and PC availability throughout the library.

Online Computer Library Center, Inc. (OCLC): It is also conducting research on mobile applications and interfaces for services that it provides in collaboration with Boopsie Inc. Boopsie is supplying the software for the mobile devices used in this study. The primary goal of this research is to collect data in order to inform and shape future mobile access to world cat.

CONCLUSION

We discussed recent developments in mobile technology and how library services have responded to these developments in this paper. One of the most recent trends that can assist libraries in meeting the high expectations of their patrons in the mobile-first culture of information consumption is responsive web design.

Libraries must promote and advocate for more mobile-friendly interfaces for many separate library systems, including the library catalogue, the link resolver, the course reserves system, licensed databases, and downloadable e-books and AV products, as a middleman between library patrons and third-party content providers such as publishers and database aggregators. Mobile technology is quickly becoming the preferred method of accessing the Internet.

The impact of mobile technology in libraries is growing in developed countries, but mobile services are being used less in India. We accepted the change but have not yet implemented it; our service is limited to SMS only. Mobile technology can be used to provide library services as an ICT option. Librarians should actively pursue the implementation of mobile technology in libraries and strive to provide quality services to their users based on their needs. Implementing this technology necessitates good infrastructure, maintenance, a budget, and skilled and knowledgeable library staff.

Library and information professionals can create mobile library websites and use them to implement various library services such as eBook selection and creation, app selection and development for library users, and circulating library materials.

REFERENCE

1. Sagar Kumbhar and Rohan Pawar(2014), **Mobile Based Services: Application and Challenge** DOI:10.13140/2.1.2373.2000
2. Abdulqawiyyu Mustapha (et.al)(2021), **Application of Mobile Technology in Information Service Delivery in Bayero University Library, Kano** Hansaben Shashikantbhai Bhoj(2020), **Mobile based library services, IP Indian Journal of Library Science and Information Technology**, <http://doi.org/10.18231/j.ijsit.2020.013>
3. Satish kumar (2015), **Excellence in Library Services by Using Mobile Technology Vol (3),11**